

# Unveiling the Black Box: Evaluating Explainable Artificial Intelligence Techniques for Convolutional Neural Network Explainability

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## INTRODUCTION

**Convolutional Neural Networks** (CNNs) excel in image classification, but their "black box" nature makes it challenging to understand and trust their predictions.

This study aims to address this by applying and analyzing three **Explainable Artificial Intelligence** (XAI) techniques: Local Interpretable Model-Agnostic Explanations (LIME), Gradient Class Activation Mapping (Grad-CAM), and Integrated Gradients (IG). By using these methods, we aim to uncover how a CNN model produces its predictions on various datasets, thus enhancing transparency and trust in AI systems.

## BACKGROUND

A **CNN** is a deep learning model designed for visual data. It uses convolutional layers to apply filters to images, capturing spatial relationships and patterns. This allows CNNs to extract features like edges, textures, and forms, essential for image recognition.

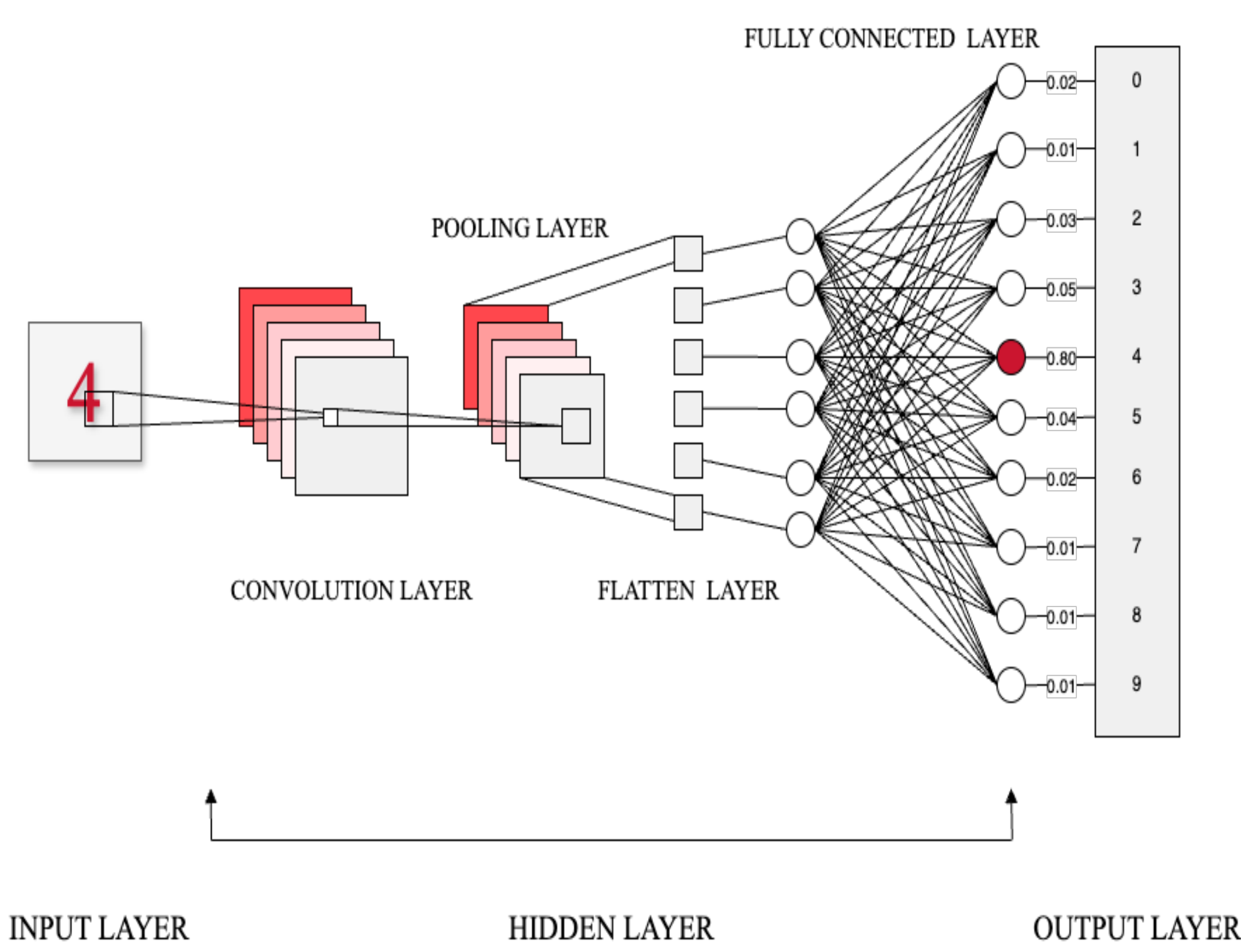


Figure 1: CNN Architecture

## DATASET

TF – Flowers dataset was used which consists of 3,670 images, which are classifies into several classes - Dandelion(0), Daisy(1), Tulip(2), Sunflower(3), and Rose(4) which are reformed to a uniform size of 224 x 224.

## METHODOLOGY

A 2D CNN was trained on the dataset in Python, consisting of three convolutional layers with ReLU activation, each followed by max pooling. The model flattens the output, processes it through a dense layer, and uses the Adam optimizer. It achieved 75.3% accuracy after 10 epochs. XAI techniques were applied to understand the model's decision-making, revealing how shapes and patterns influence predictions across the dataset.

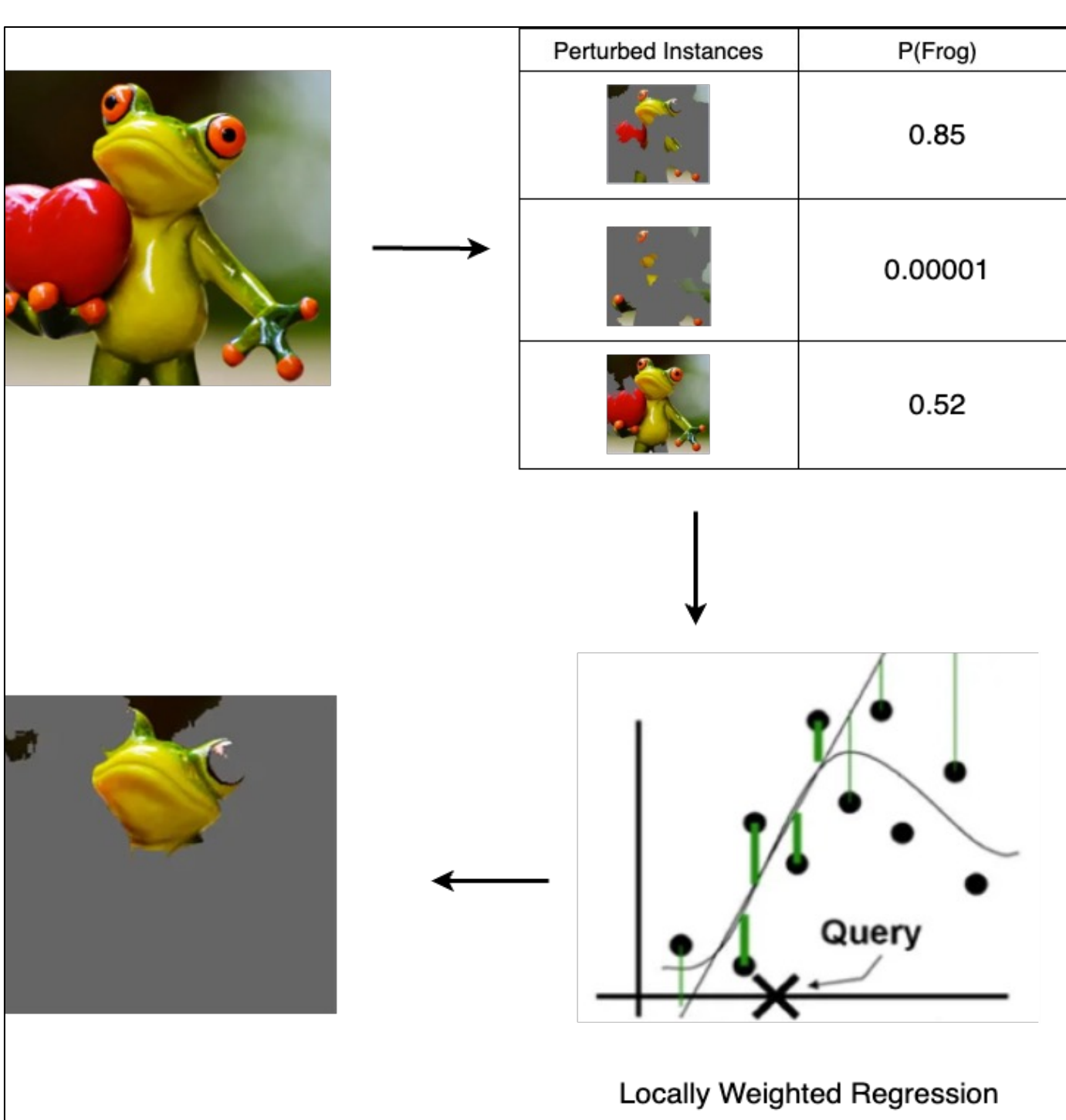


Figure 2: LIME Architecture

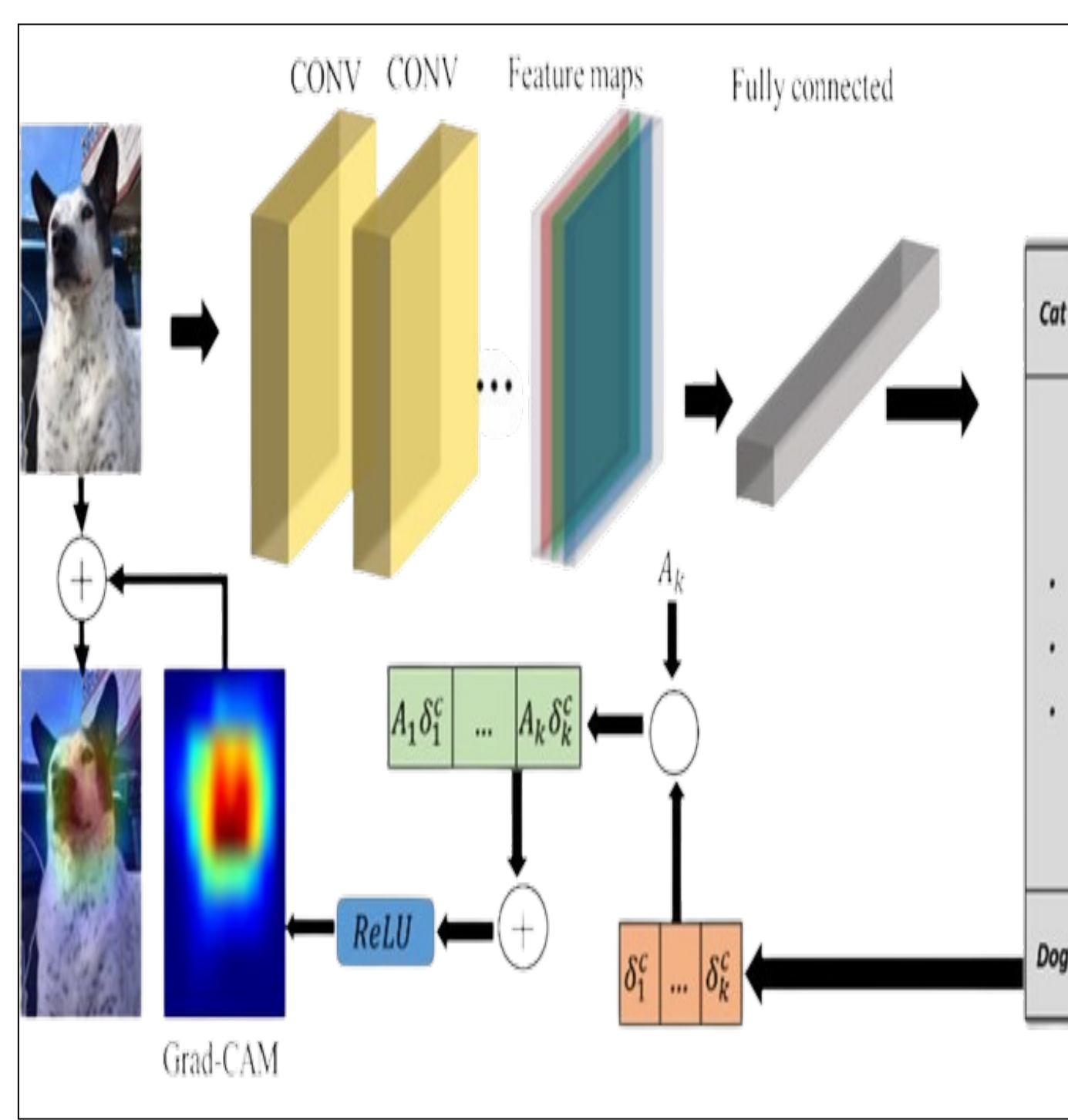


Figure 3: GRAD-CAM Architecture

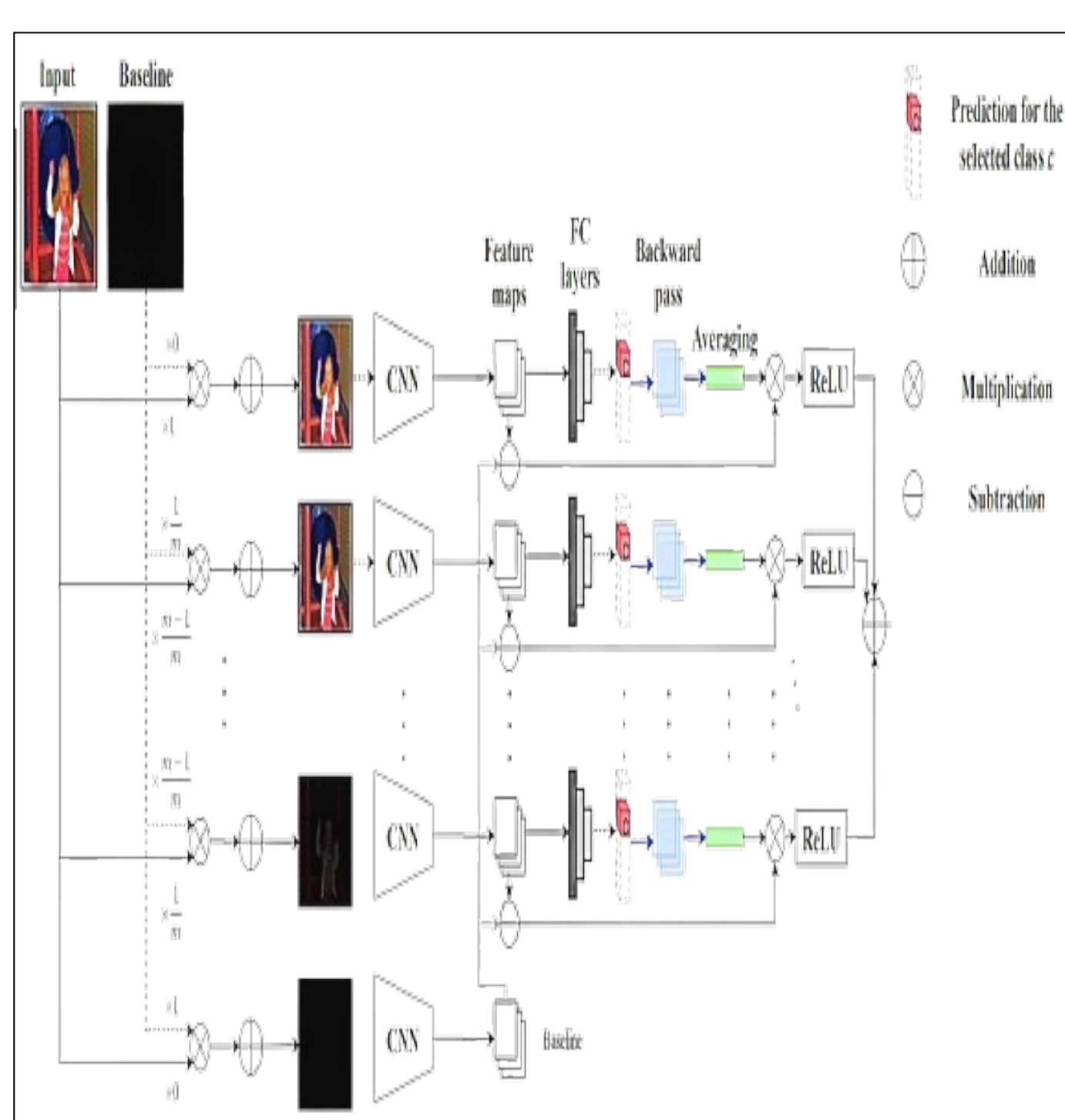


Figure 4: IG Architecture

## RESULTS



Figure 5: Original Image



Figure 6: LIME Explanation

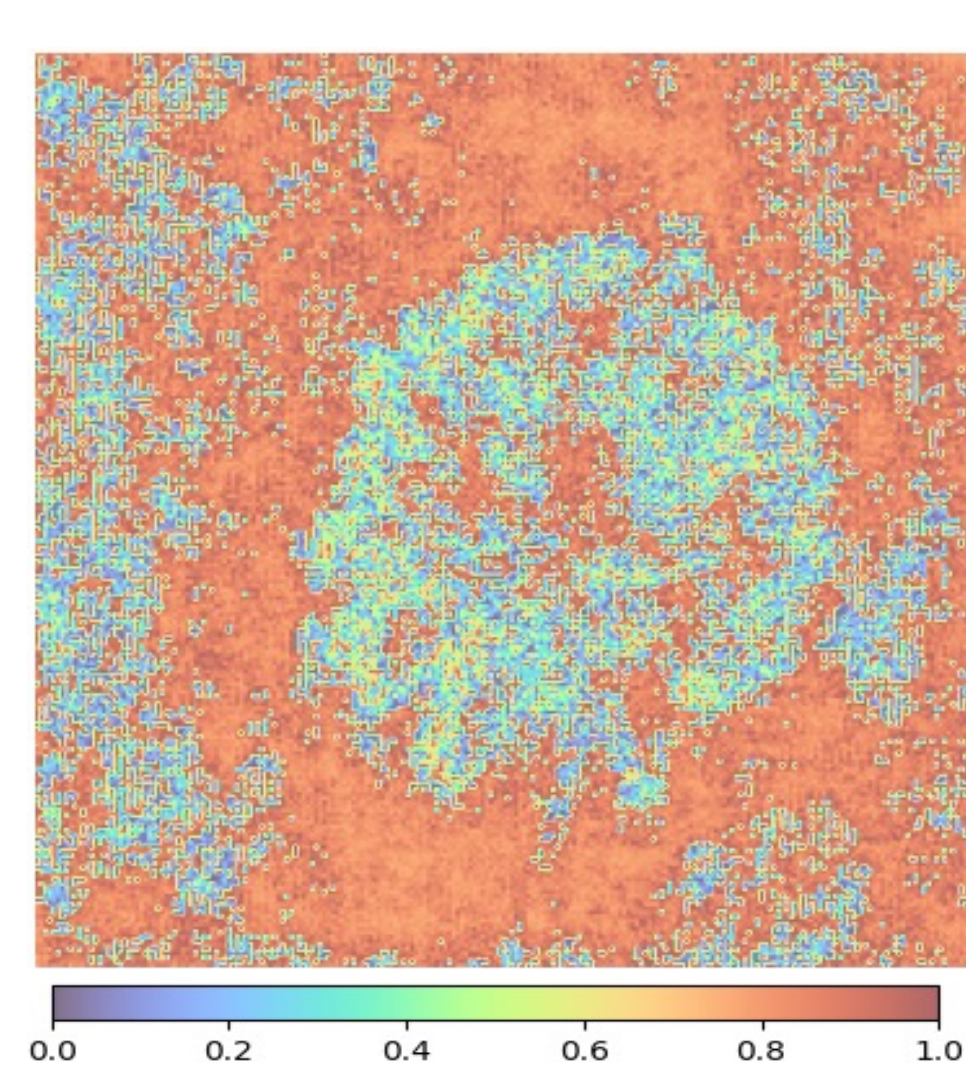


Figure 7: IG Explanation

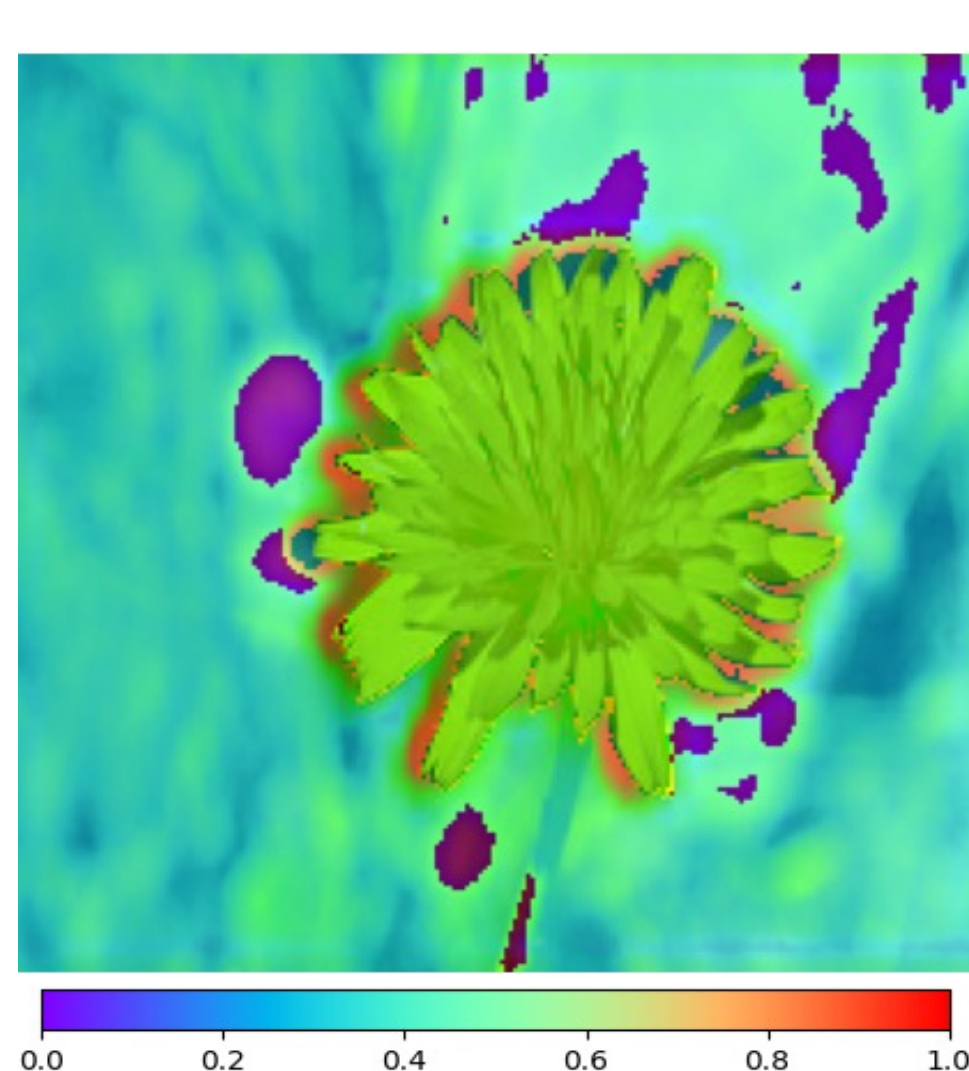


Figure 8: Grad-Cam Explanation



Figure 9: LIME Explanation of a False Prediction



## CONCLUSION

**Grad-CAM** highlights influential flower patterns, providing visual insights into the areas impacting predictions. **LIME** offers local explanations by focusing on fine-scale details, enhancing transparency for specific predictions. **Integrated Gradients** emphasize each pixel's contribution, complementing the other methods.

Together, these XAI techniques boost model trustworthiness, crucial for applications like flower disease detection that require accurate and interpretable AI solutions. By combining these methods, researchers can achieve comprehensive explanations of CNN predictions, ensuring robustness and reliability in real-world scenarios.

## FUTURE WORK

Our future research will concentrate on utilizing XAI techniques to improve our comprehension of hyperspectral image classification. These techniques will be implemented on a CNN created to detect different forms of bacteria. With the help of these XAI tools, our goal is to shed light on how the CNN makes decisions, offering explanations for its predictions. This method will enhance both model transparency and confidence in AI applications, particularly in crucial areas such as healthcare.

## REFERENCES



## ACKNOWLEDGEMENTS

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